Manages

Computer Science & IT

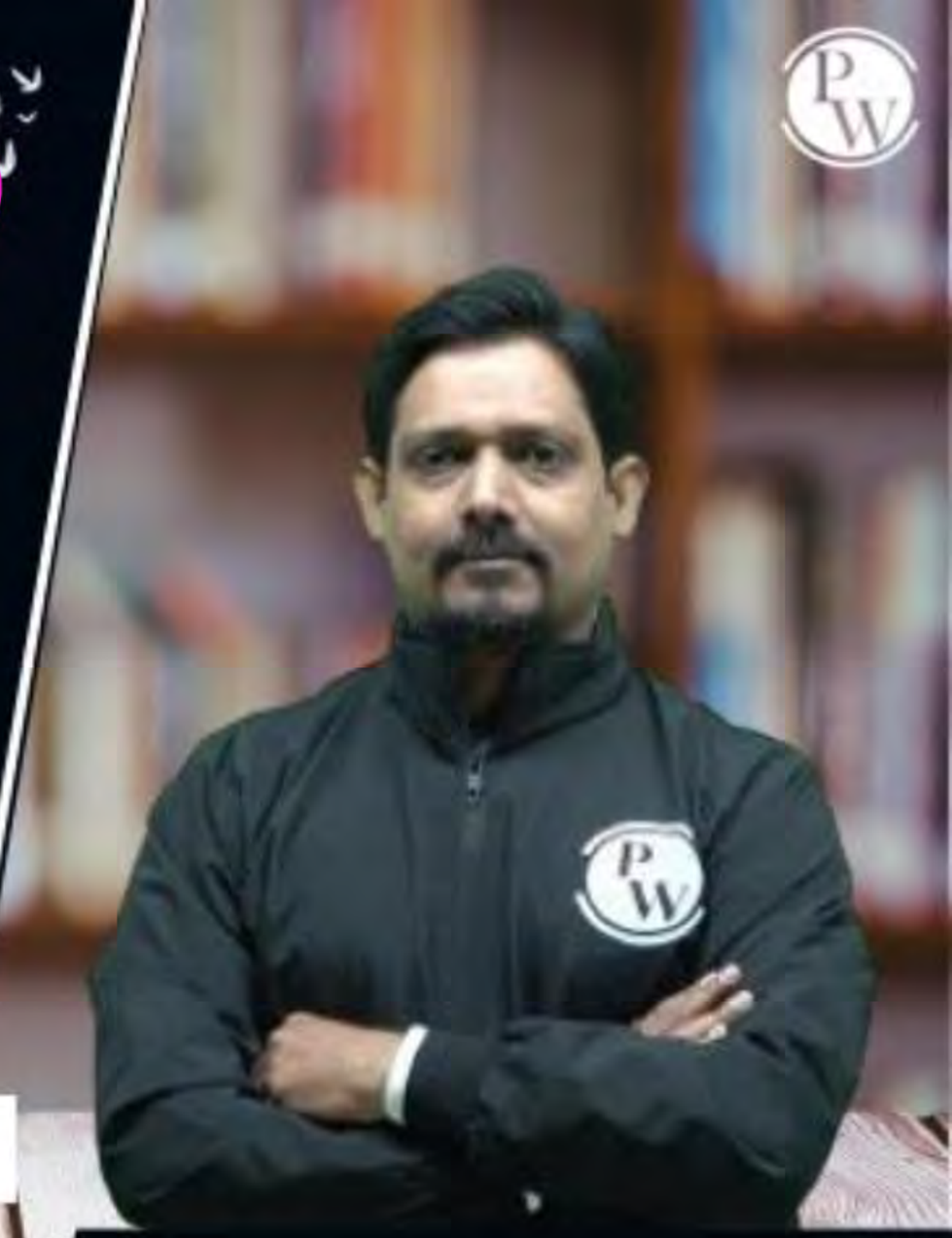
1st year



Data Structure & Programming

Array 04

Lecture No. 05



By- Abhishek Sir

Recap of Previous Lecture



Topic — Lower Triangular Matrix Column major Order

Topic — Upper Triangular Matrix

Home work

Topic

Topic

Topic

Slide

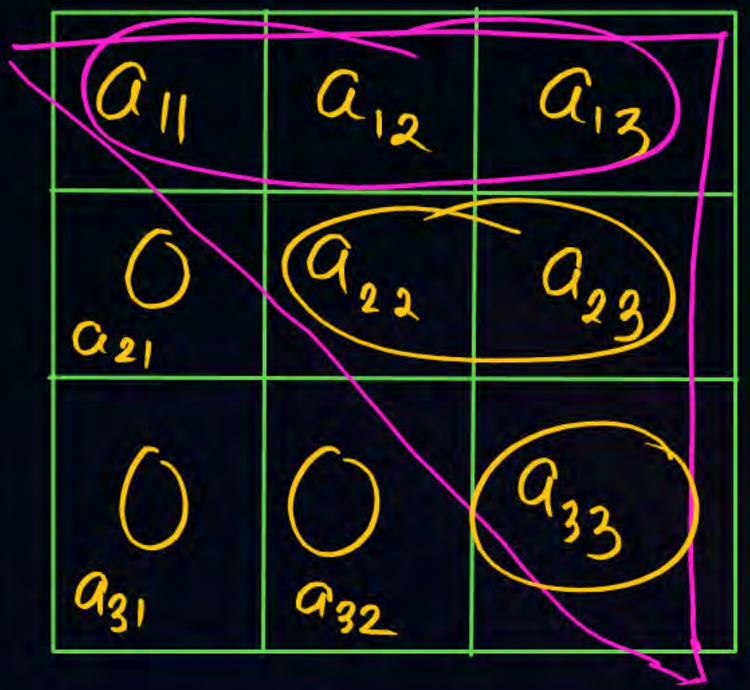
Topics to be Covered



- Topic: 3-D matrix ✓
- Topic: Tridiagonal Matrix ✓
- Topic: Practice Problem
- Topic: Upper Triangular matrix. Homework
- Topic

Slide

Topic : Upper Triangular Matrix



Elements below principle diagonals are zero

Row major order

A[i][j] 1st Row

$i > j$ A[i][j] = 0

Slide

Topic : Upper Triangular Matrix

Non zero elements are stored

Elements below principle diagonals are zero

Row major order

a_{11}	a_{12}	a_{13}
0 a_{21}	a_{22}	a_{23}
0 a_{31}	0 a_{32}	a_{33}

A[i][j]

1st Row — n

2nd — n-1

(i-1)th — n-(i-2)

} (i-1)

$i > j \quad A[i][j] = 0$

$$BA + \left[n(i-1) - \frac{(i-1)(i-2)}{2} + (j-i) \right] \times Size$$

Slide



Topic : Upper Triangular Matrix

UTM

a_{11}	a_{12}	a_{13}
0 (a_{21})	a_{22}	a_{23}
0 (a_{31})	0 (a_{32})	a_{33}

Elements below principle diagonals are zero

Column major order:

$$BA + \left[\frac{j(j-1)}{2} + i - 1 \right] \times Size$$

$i > j \quad A[i][j] = 0$

Slide



Topic : Triangular Matrix

Lower Triangular Matrix : RMO :

$$BA + \left[\frac{i((i-1)}{2} + j - 1 \right] \times Size$$

CMO :

$$BA + \left[n(j-1) - \frac{(j-1)(j-2)}{2} + (i-j) \right] \times Size$$

Upper Triangular Matrix RMO :

$$BA + \left[n(i-1) - \frac{(i-1)(i-2)}{2} + (j-i) \right] \times Size$$

CMO :

$$BA + \left[\frac{j(j-1)}{2} + i - 1 \right] \times Size$$

Slide

Topic : 3-D Matrix (C – Language)



BA = 100

Size : 2B

A [1..3] [1..2] [1,2]

Row-1 A [1..3] [1..2] [1..2]

3 frame Row Column

frame

frame-3

9	10
11	12

frame-2

5	6
7	8

frame-1

1	2
3	4

3 × 2 × 2 = 12 No. are present

A [3] [2] [1]

frame

frame 1

frame

frame

120

100	102	104	106	108	110	112	114	116	118	120	122
1	2	3	4	5	6	7	8	9	10	11	12

Row-1 Row-2 ..... Row-1 Row-2

Slide

Topic : 3-D Matrix (C – Language)

$A[3][2][1]$

$i \quad j \quad K \quad 2$

$BA = 100, \text{ Size} = 2B$

$$100 + [(3-1) \times 2 \times 2 + (2-1) \times 2 + (1-1)] \times 2$$

$(J - LB_2)$ $\qquad$ $(K - LB_3)$

$A[1..3][1..2][1..2]$

$UB_2 - LB_2 + 1$

$UB_3 - LB_3 + 1$

$= 100 + [8 + 2 + 0] \times 2$

$= 100 + 20 = 120$

Slide



Topic : 3-D Matrix (C – Language)

formula : $A[Lb_1 \ldots ub_1][Lb_2 \ldots ub_2][Lb_3 \ldots ub_3]$

Address $A[i][j][k]$

$Lb_1 = Lb_2 = Lb_3 = 0$

C Language

$$BA + [(i - Lb_1)(ub_2 - Lb_2 + 1)(ub_3 - Lb_3 + 1) + (j - Lb_2) \times (ub_3 - Lb_3 + 1) + (k - Lb_3)] \times Size$$

Rows — Column — Column

Slide



Topic : 3-D Matrix (C – Language)

Q A[1..100][1..100][1..100]

BA : 1000 Size : 2B

what is the Addrss of A[50][49][50]

Row major order?

$1000 + [49 \times 100 \times 100 + 48 \times 100 + 49] \times 2$

$1000 + [490000 + 4800 + 49] \times 2 = 1000 + 494849 \times 2$

float a : 2.0;

4 Byte

989698
1000
990698

Slide

Topic : Tridiagonal Matrix



a_{11}	a_{12}	0	0	0
a_{21}	a_{22}	a_{23}	0	0
0	a_{32}	a_{33}	a_{34}	0
0	0	a_{43}	a_{44}	a_{45}
0	0	0	a_{54}	a_{55}

3 diagonal are present

above main diagonal

main diagonal - Non zero

Non zero elements present in Main diagonal, above main diagonal and below main diagonal

5x5 Square matrix

Below main diagonal

Slide

Topic : Tridiagonal Matrix

a_{11}	a_{12}	0	0	0
a_{21}	a_{22}	a_{23}	0	0
0	a_{32}	a_{33}	a_{34}	0
0	0	a_{43}	a_{44}	a_{45}
0	0	0	a_{54}	a_{55}

3 diagonal are present

5×5 - main diagonal :- 5

above & below main diagonal. No. of elements

4 above 4 below

if n×n matrix : main diagonal : n

above : n-1 , below n-1

total NonZero :- n + n-1 + n-1

= (3n-2)

Slide

Topic : Tridiagonal Matrix

(j-(i+1))

a_{11}	a_{12}	0	0	0
a_{21}	a_{22}	a_{23}	0	0
0	a_{32}	a_{33}	a_{34}	0
0	0	a_{43}	a_{44}	a_{45}
0	0	0	a_{54}	a_{55}

Nonzero element are stored one after another in * Row major order

100	102	104	106	108	110	112	114	116	118	120	122	124
a_{11}	a_{12}	a_{21}	a_{22}	a_{23}	a_{32}	a_{33}	a_{34}	a_{43}	a_{44}	a_{45}	a_{54}	a_{55}

Row-1 Row-2 Row-3 Row-4 Row-5

Address of A[i][j]: ith row (i-1) rows are Complemted

(i-1) rows consist of No. of element $3\times(i-1)-1$

$j-i+1$

$2-3+1=0$

a_{43}

$i\ j$

Slide



Topic : Tridiagonal Matrix

A[i][j] : Before ith Row (i-1) Rows are arranged

a_{11} i=1, j=1

$3(i-1)-1$

in ith Row : $j-i+1$

$BA + [3(i-1) - 1 + j - i + 1] \times Size$

100 + [2+1-3] x2

100 + 0 x2

= 100

$BA + [3i - 3 + j - i] \times Size = \boxed{BA + [2i + j - 3] \times Size}$

Slide



Topic : Tridiagonal Matrix

$$BA + [2i + j - 3] \times Size$$

a_{55} :

$$100 + [2 \times 5 + 5 - 3] \times 2$$

$$100 + [12] \times 2 = 124$$

Slide



Topic : Question

#Q Consider the LTM consists of 45 non zero elements. If The dimension is n*n then the value of n is 9.

$1+2+3+\cdots+n$

$\frac{n(n+1)}{2} = \frac{9\times10}{2} = 45$

$n = 9$

$n = 10$

$\frac{10\times11}{2} = 55$

Slide



Topic : Question

#Q If natural number are stored in Array A[-10..10][20..30]

Then the number stored in A[5][15] is [RMO]

(*)

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

$$1 + [(5-(-10)) \times (30-20+1) + (25-20)] \times 1$$

$$1 + [15 \times 11 + 5] \times 1$$

$$1 + [165 + 5]$$

$$= 171 \checkmark$$

A[1..4][1..4]

$$1 + (12 + 2) = 15$$

A[3][4]

$$1 + (2 \times 4 + 3)$$

$$1 + 11 = 12$$

Slide



Topic : Question

Home work

#Q Consider the LTM a[-4..4][0…8].Non zero elements are stored in row major order. What is the address of A[0][3]. Base address is 1000 and size of each element is 2B.



Topic : Question

Q Assume that Tridiagonal Matrix of order (n × n) is mapped into one dimensional array using row major order then calculate the location of an element a[i,j]. (L0 is base address and size is 1 Byte)

A. L0 + (2i – j – 3)

B. L0 + (2i + j + 3)

C. L0 + (2j + i - 3)

D. L0 + (2i + j - 3)



Topic : Question

Homework

#Q Assume that Tridiagonal Matrix of order $(n \times n)$is mopped into a one dimensional array using column major order, then calculate the location of an element a(i,j)(L0 is base address and size is 1 Byte)?

(A) $L_0+(2j+i-3)$

(B) $L_0+(2i+j-3)$

(C) $L_0+(2i-j-3)$

(D) $L_0+(2j+i+3)$

Slide



2 mins Summary

Topic

Topic

Topic

Topic

Topic

3-D array

Tridiagonal Row major order

Slide

THANK - YOU

Slide